BEHAVIOUR AND MANAGEMENT OF POULTRY

By

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- **The term poultry covers feathers wide of birds which include dead or dressed and living birds. The living includes chickens, turkeys, ducks, geese, swans and pigeons.**

Chicken

Turkey

Ducks

GEESE

SWAN

Pigeon

Ostrich

Quail

Importance of poultry:

1. Act as a source of income.

2. used as a food, so the meat and eggs are used for human food.

3. in industrial uses:

- Fertile eggs used for vaccines preparation.
- Inedible eggs used for animal food and fertilizers.
- Egg white used in Pharmaceutical preparation, paints and as adhesives.
- Egg yolk used for cake making, soap and shampoos.
- Egg shell used in making mineral mixture and in fertilizers
- Feathers act as animal feed.
- Endocrine used for biological products.

Importance of poultry:

4. in research due to:

- a. Chicken is more sensitive than rates
- b. Cheap and available.
- c. Large number of them can be hatched at the same time.

POULTRY BEHAVIOR:
1. INGESTIVE BEHAVIOR:

- Chickens and turkeys ingest their feed by pecking; ducks scoop their feed with their broad soft bills'. Except geese (which graze on grass), poultry do not eat much forage. poultry is governed by the energy concentration of the feed.

- **Chickens:**
- Chicks do not peck much until their second day after hatching presumably due to ingestion of the yolk sac during hatching. Normal pecking experience requires some light.
- Initially, chicks peck like small hammer and ingest both nutritive and non-nutritive substances.

Ingestive behaviour is influenced by:

- 1. Dominance order: The relationship between birds in the flock where **the dominant bird feeds more than inferiors**. Dominant reducing the feed consumption of those in the lowest rank.

- 2.Hungry: Where the hunger birds eats more than others.

- 3. Feeding space: As the birds grow up the feeding space should be increased.

- 4. Group size: They peck and fed more in the presence of feeding companions.

Ingestive behaviour is influenced by:

- 5. Energy content: Birds eat to satisfy their energy requirements.
- 6. Time of day: Birds eat much during morning and late afternoon before sunset.
- 7. Age of bird: Older bird eats more than young one.
- 8. Sex of birds: as the producer females eats more than males.
- 9. Breeds of Birds: There were a difference between layers and broilers in their feed intake.

TURKEY:

- Turkeys swallow without the necessity of raising the head.

- Drinking is accomplished differently, after getting a beak full of water, the bird raises its head, extends the beak upward and repeats several rapid closures of the beak.

- Newly hatched poults pick at bright spots and small objects that contrast with the background. Thus, they may be encouraged to consume feed and water by the use of colored feed, or by placing brightly colored marbles in the feed and waters.

Ducks:

- **Ducks employ several feeding methods, including :**

1. Dabbling of the bill along the water surface or in mud.

2. Feeding from the bottom of ponds.

3. Snapping flying insects by rapid closing of the bill.

- Drinking by ducks involves a distinctive movement sequence; the bill is dipped in water and then lifted slightly above the horizontal.

QUAIL:

- **The ingestive behavior of the quail is characterized by:**
- 1. Sharp increases in food intake shortly after the onset of **morning light and during the period of approximately 3 hours prior to darkness**.
- 2. Preference for **sweet and sour** solutions, and avoidance of **salt and bitter** solutions.
- 3. The ability to tolerate **moderate levels of salt in the drinking water** for prolonged periods, although they normally avoid a salt solution.

2.ELIMINATIVE BEHAVIOR:

- In mammals, urine and feces are voided separately; but in poultry the end products coming from the urinary and gastrointestinal tracts and mixed together and voided by the cloacae.

- Poultry manure has higher nitrogen content than manure from four-footed animals. Chickens deposit their excreta at random.

3.COMFORT AND GROOMING BEHAVIOR:

- Comfort movements of birds composed of: Stretching, yawning, preening, dust bathing and shaking.

1. The stretching has two types bilateral in which the wings are half opened and stretched upward and forward or unilateral.

2. Yawning: may occur in conflict situations.

3. Preening: The total time spent preening by adult fowls is 2-4 % of 24 hours. The frequency of preening varied with the time of the day and with the frequency of sleeping.

3.COMFORT AND GROOMING BEHAVIOR:

4. Dust bathing is a highly social event in the fowl in that a dust-bathing bird is usually joined by others, but birds, in battery cages may perform it. During dust-bathing the bird may perform bill raking, vertical wing-shaking, scratching, lying on the side, and head rubbing.

5. Feather ruffling is another form of comfort movements.

6. Head shaking and tail shaking.

4.RESTING BEHAVIOR:

- Sleep in fowls is partly independent of light.
- The activities of birds depended not only on endogenous periodicity but also on synchronized external factors, one of which is the light intensity which varies with the season. Light plays a subsidiary role in the activity rhythm of fowls.

5.REPRODUCTIVE BEHAVIOR:

- Reproduction is the first and most important requisite of livestock breeding. Without young being produced, the other economic traits are of academic interest only. Thus, it is important that all those who breed animals should have a working knowledge of sexual behavior.

- Sexual behavior involves **courtship and mating**. It is **largely controlled by hormones**. Each animal species has a special pattern of sexual behavior. As a result, interspecies mating do not often occur.

CHICKENS:

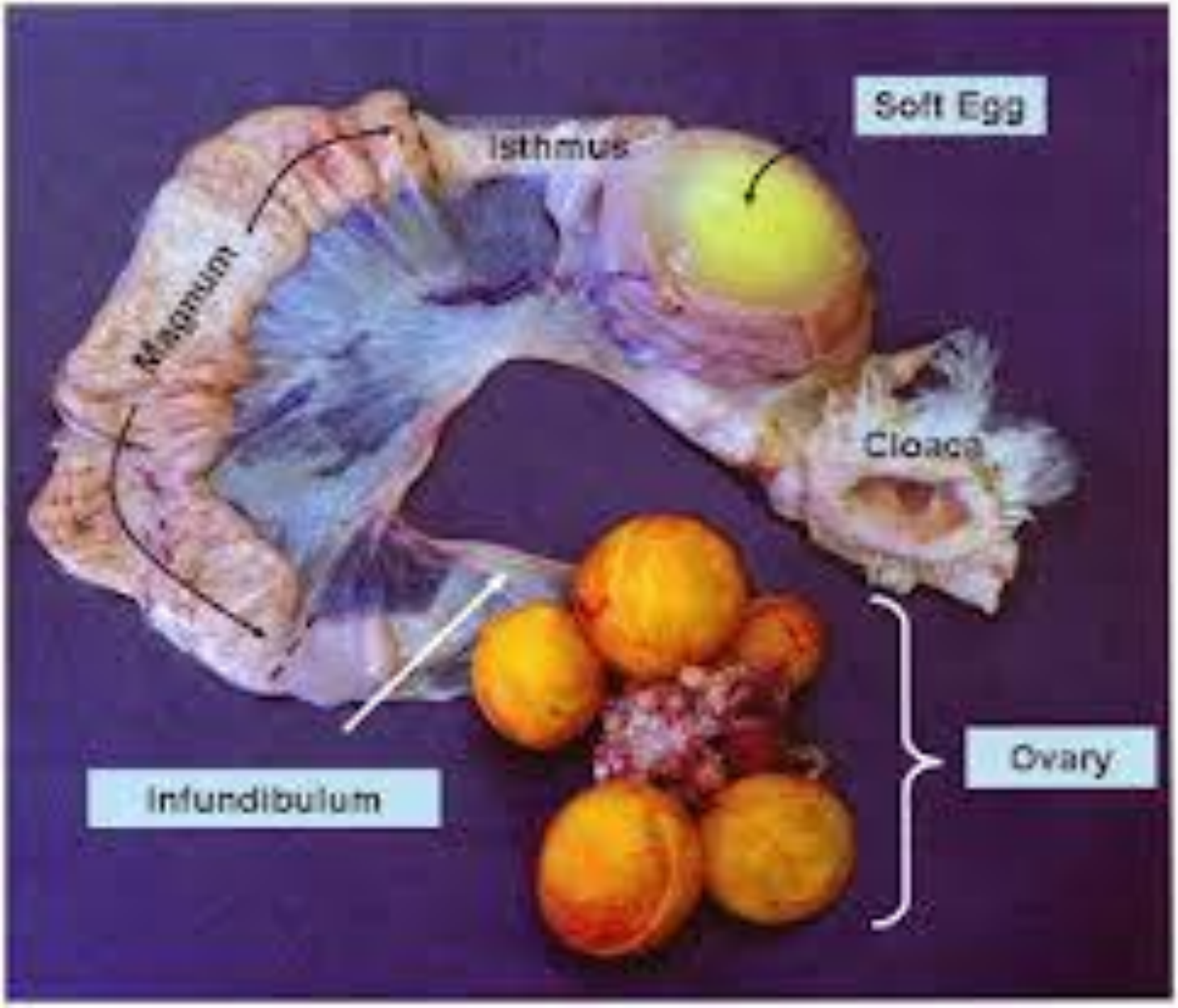
- In chickens, which are polygamous, sexual behavior is usually referred to as mating behavior. Mating in chickens is preceded by various behavior patterns known as displays or courting, which synchronize sexual activities of males and females.

- **Sexual maturity** occurs in chickens at about 22-26 weeks.

- **Stocking rate** one male for 10-12 chickens in natural mating system.

- **Sperm fertilizing capacity** for 32 days after insemination.

- **Ovulation:** in chicken normally occurs after 30 minutes from the egg laying. The average interval between two successive eggs laid in the same clutch is about 27 hours.


Soft Egg
Isthmus
Magnum
Cloaca
Ovary
Infundibulum

Chickens:

- **Laying of the eggs:** Eggs are normally formed with the small end first as they move through the oviduct. When the large end appears first indicate moist egg.

- **Clutch:** Is the number of the eggs laid by a hen on consecutive days.

- **Molting:** At the end of eggs laying season, molting occurs which characterized by: 1. Losses of feather and growing of a new one and 2. occurring in females only.

Clutch

Molting

- **The courtship behavior of the cock may be expressed in one of the following forms:**
- 1. Waltzing with wing fluttering or dancing
- 2. Extending head and grasping hen's comb or Hackle
- 3. Chasing hen, followed by grasping her comb or hackle and the hen may respond by:
- 1. Negatively, which may be accompanied by vocalization, panging from faint screams to loud squawks, by:

    a. Stepping aside.

    b. Walking or running away.

    c. Struggling.
- 2. Positively, by crouching, often with head low and wings spread.

MALE *FEMALE*

Sexual approach → **1 Escapes**
2 Avoids

Mounts and treads ← **3 Sexual crouch**

→ **Moves tail to one side and everts cloaca**

Spreads tail and everts cloaca ←

→ **Vents meet**

Ejaculates and steps off ←

→ **Stands and shakes**

- After courting the mounting occur by grasping the cock to the comb or hackle of the hen and treading, the hen moves her tail to one side and everts cloacae then the cock rears up and everts cloacae till vents meet with ejaculation after that the cock steps off in forward direction or waltz in circle while the hen stands and shakes feather or may run.

- **<u>Factors affecting sexual behavior in chickens:</u>**
- 1. Both early **social experience and hormonal level** influence the level of sexual behavior.
- 2. The **earlier the separation of males from females**, the less the sexual performance of males.
- 3. The relation of **sex hormones** and sexual behavior is well established.
- 4. **When several cocks** are placed with a flock of hens, the **dominant cock** (the one ranking highest in the peck order) is usually most successful in mating, fertilizing a large number of eggs, and siring the most chicks.

- **<u>Factors affecting sexual behavior in chickens:</u>**

- 5. Dominance among hens interferes with mating, because such hens do not readily submit to the crouch.

TURKEYS:

- The mating behavior in turkeys tends to follow a chain reaction similar to chickens. However, the following differences exist between the sexual behavior of chickens and turkeys.

- 1. The movements of chickens during courtship and mating are **more rapid and feather display of the male is less elaborate than in turkeys**.

- 2. Typically, male chickens force mating with hens, but this behavior is not seen in turkeys.

TURKEYS:

- 3. **Turkey's** female sometimes **follows** the male and crouches near him, whereas female chickens seldom approach the cock before crouching.

- 4. If another male attempts to mount, cockerels may treated each other whereas male turkeys will not.

Ducks:

- Ducks form pair-bonds. Both males and females breed in their first year; and normally they raise only one brood annually. Unlike most birds, male of water-fowls have a well-developed penis.

- Social courtship displays and associated with pair formation occur only during the autumn, winter, and early spring months.

Female sexual displays include the following:

- 1. **Inciting** (mate preference): Inciting appears to indicate the female's preference for one male and the rejection of others. a distinctive display associated with pairing and maintenance of the pair-bond, in which the female walks, swims, or flies beside or behind a chosen male, while making threatening movements over one shoulder and giving loud calls.

Female sexual displays include the following:

- 2. **Nod-swimming** associated with social court-ship in which the female swims rapidly among the males with her head stretched forward over the water surface. Head nodding in which the head is held high but moved forward and backward, occurs in similar situations, both actions stimuli males to perform displays.

Female sexual displays include the following:

- 3. **Repulsion**, characteristic of incubating or brooding females when they are harassed by drakes intent on rape, in which the head is tucked back, the bill is opened, the plumage on the back and flanks is ruffled, the tail is fanned, and loud, harsh calls are emitted.

- 4. **Distraction displays**, given when incubating females are flushed from the nest. The female flaps across the ground or water, thrashing with both wings and giving loud quacks.

Male Sexual displays include the following:

- 1. **Preliminary displays**, The male adopt a characteristic posture with the head sunk in the shoulders and the head feathers ruffled, followed by head-shakes, head-flicks, prolonged tail-wags, and swimming shakes.

Male Sexual displays include the following:

- 2. **Major courtship displays**, characterized by:

- a. **Grunt-whistle**, characterized by the lowering of the bill to the water surface; arching the body upward; flicking the bill to one side and sending a fine spray of water toward the female; and uttering a loud whistle followed by a grunt.

- b. **Head-up, tail-up**, a complex displays involving a sudden rising of the head along with simultaneous vertical cocking of the tail and rising of the closed wings. The movement is accompanied by a loud whistle. As wings and tail are lowered, the male turns his head so as to point his bill toward a female (the mate in the case of paired birds).

Male Sexual displays include the following:

- c. **Down-up**, involving a sudden complex of body actions in which the drake dips his breast deeply into the water; jerks his bill upward and outward, flipping up a column of water, and emits a whistle when the head is at the highest level.

Male Sexual displays include the following:

- 3. **Ritualized jump-flights,** in which, in rapid succession, the drake rises steeply with head extended horizontally, flies 3 to 10 ft, alights a short distance in front of the female, and drinks.

- 4. **Bridling**, in which the drake pulls his head back so that the breast is raised slightly from the water. This display normally follows copulation.

Reproduction in fowl:

- **Development of chick:**
- **After ovulation**, the yolk enters the Infundibulum, the sperm moves toward the egg for fertilization. The fertilization occurs at the germinal disc on the yolk. The shell and the membranes protects the embryo from bacteria and moulds, also the white (Albumin) act as bacteriostatic.

- **In the event of the fertilization**, the embryo begins to develop around this well define germinal disc, this area is visible and seen by naked eye when the eggs broke.

Reproduction in fowl:

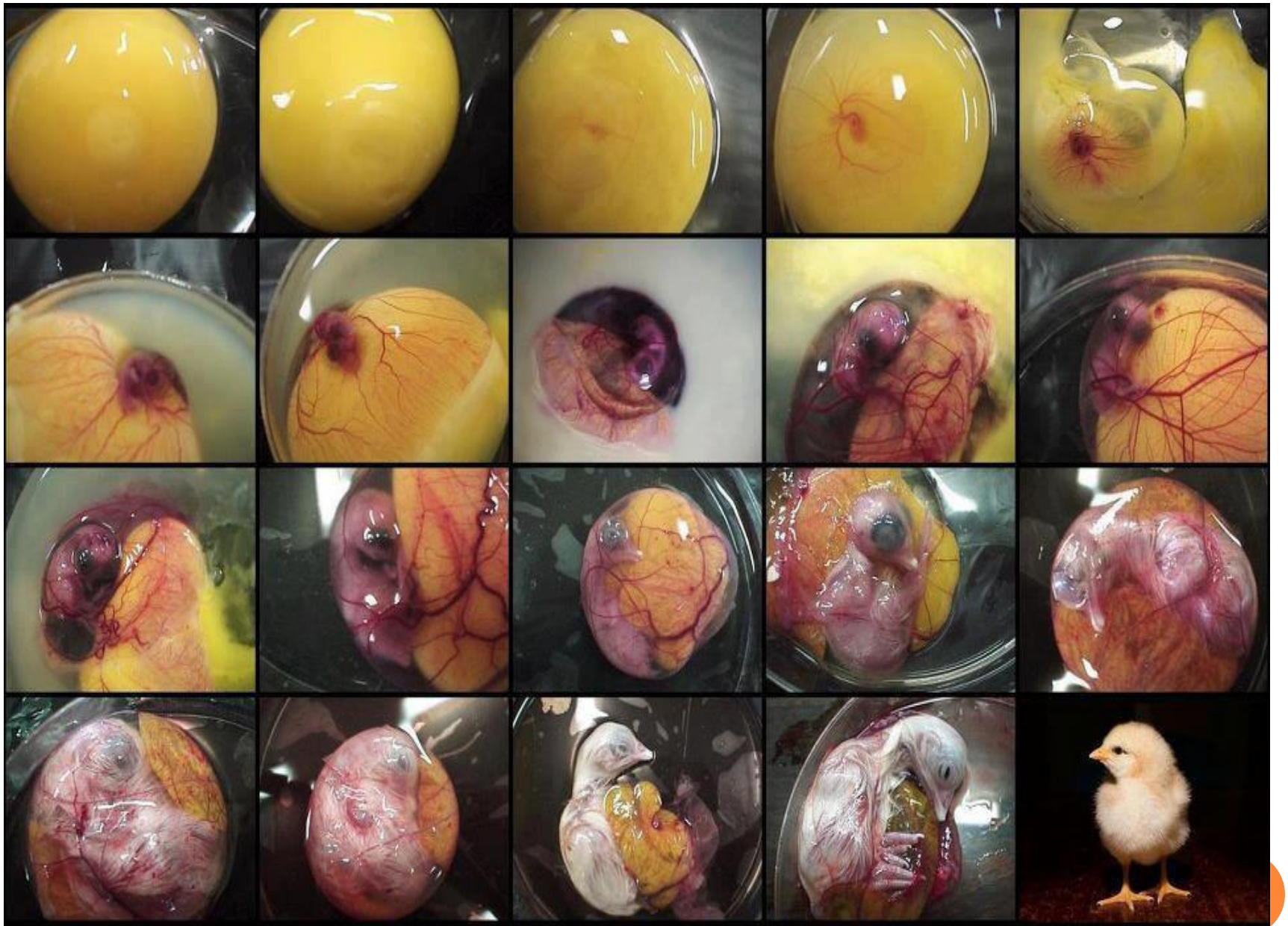
- **Development of chick:**
- **Within 48 hours**, the embryo as settled of **blood circulation** between the embryo and sustaining yolk. So no placenta present here in contrast to the other mammals and the interacted blood vascular net work used for:

a. Carry the necessary nutrient for feed.

b. removal of waste products At the end of the third days the embryo has a full complement of the membranes (allantois, chorion and amnion).

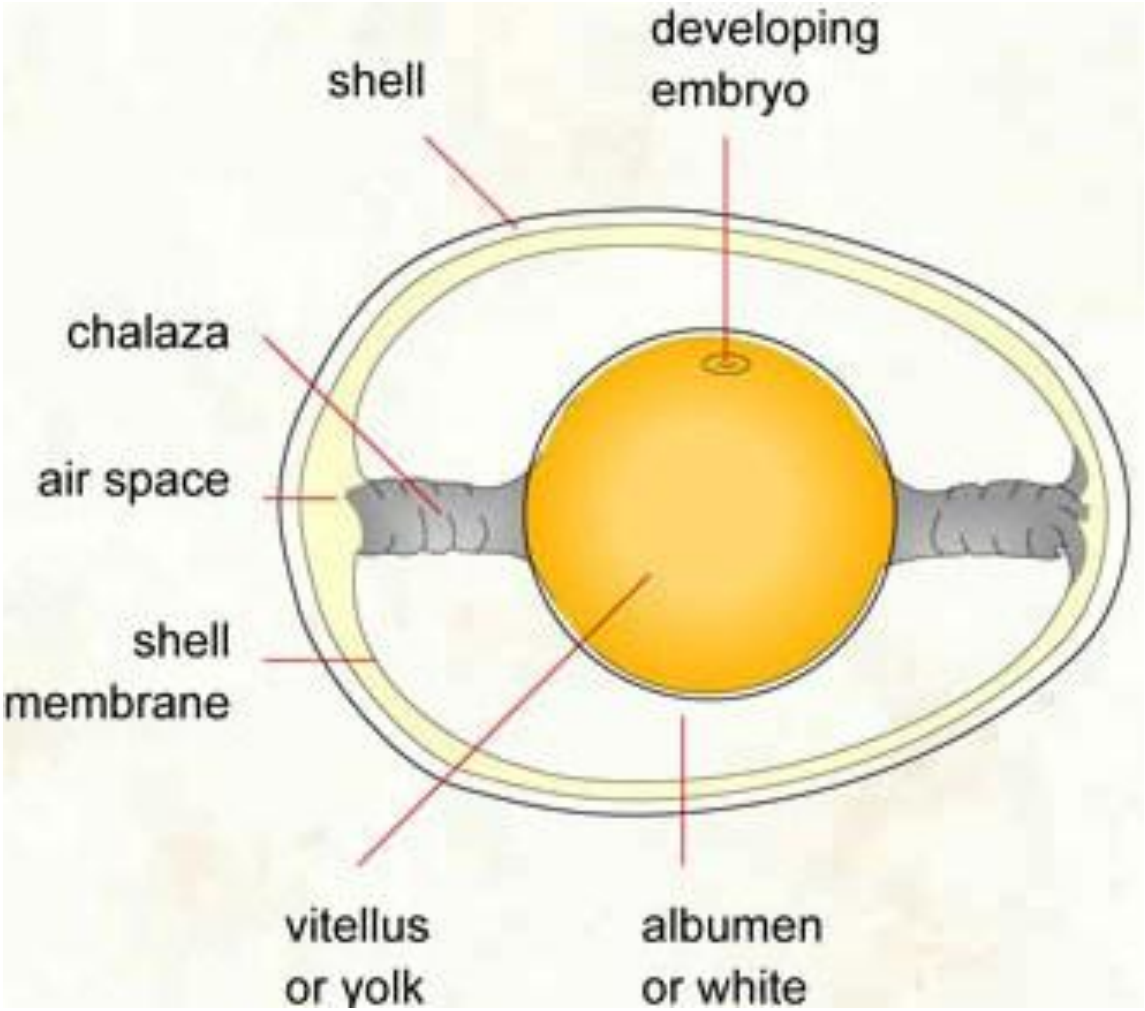
Reproduction in fowl:

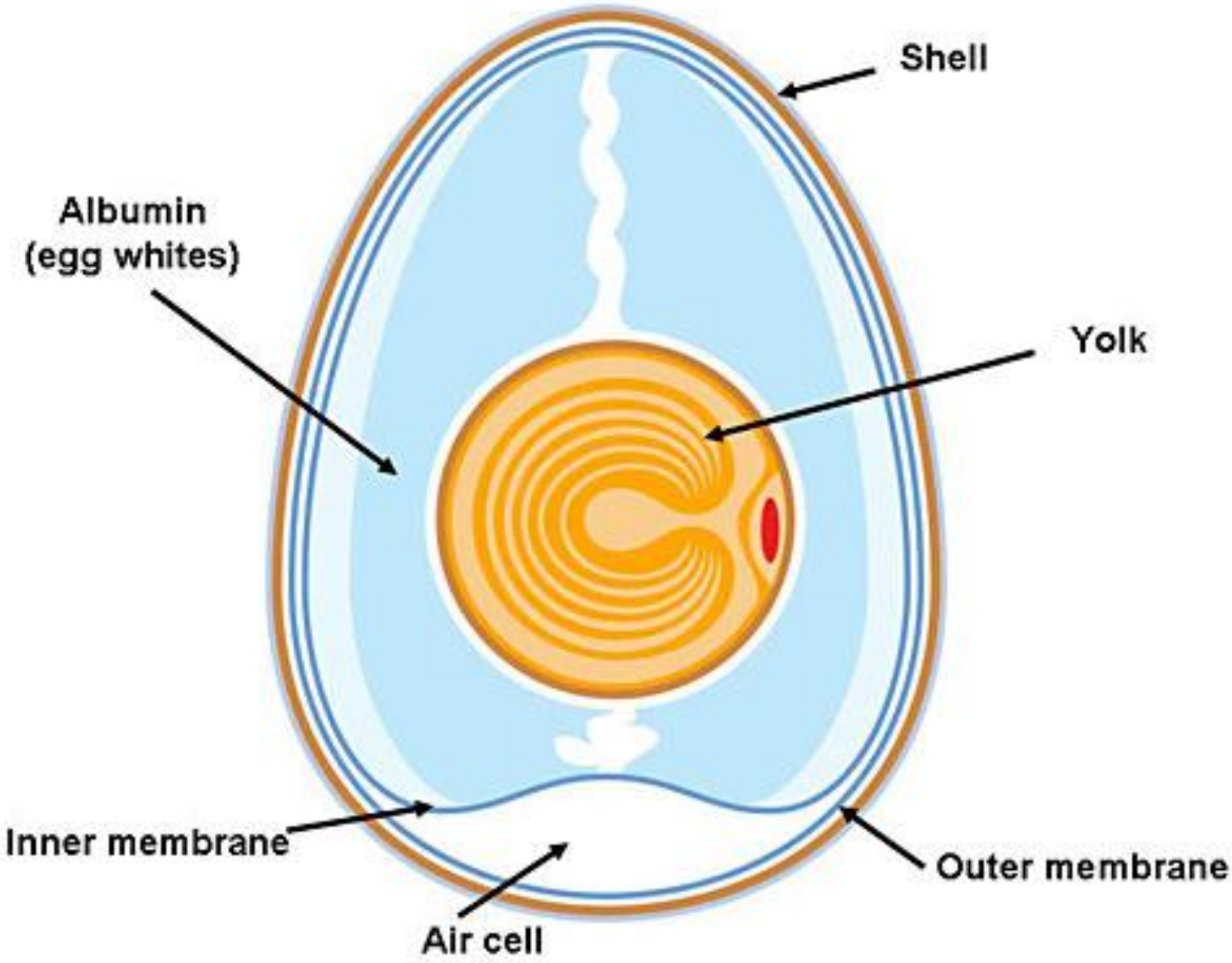
- **Development of chick:**
- **At about the midway through the incubation period**, the embryo assumes the normal position, which is lie on its left side along the longest axis of the eggs, and then head is tucked under the right wing facing the large end of the egg.

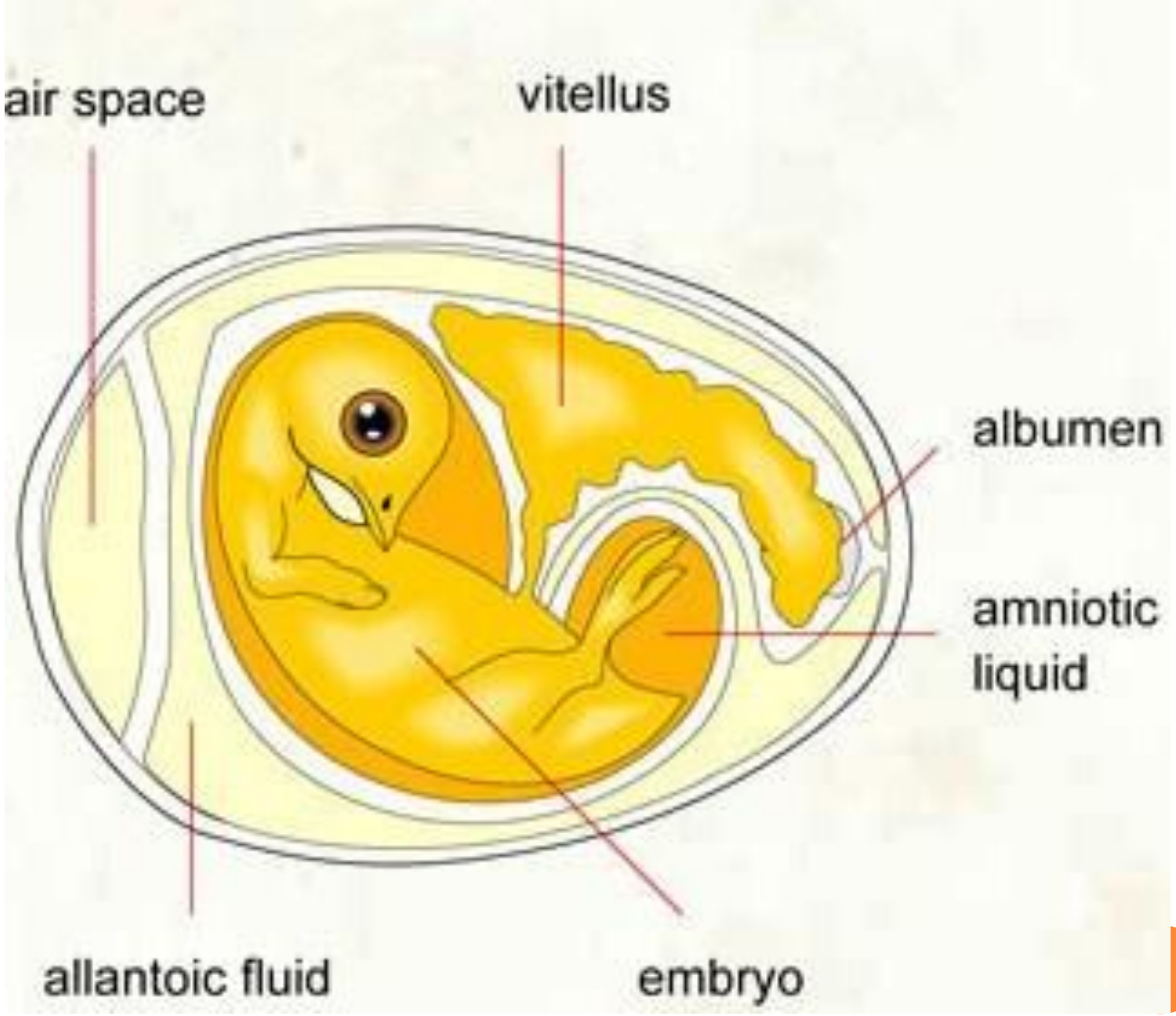


VET CRESCENT SOCIETY

Structure of egg




Shell
Albumin
(egg whites)
Yolk
Inner membrane
Outer membrane
Air cell


air space
vitellus
albumen
amniotic liquid
allantoic fluid
embryo

14
15
16
17
18
19
20
21

INCUBATION OF EGG:

When calculating the number of eggs which must be incubated to produce a given number of laying pullets. we must know that:

- 1. only 50% of the chicks hatched will be pullet,
- 2. only 60 - 70% of the eggs will probably hatch
- 3. about 10% of the pullets will have to be culled. Thus three to four eggs must be incubated for every laying pullet required.

Incubation of eggs carried out by:
A. Brooding Hen:

- At the end of clutch hen usual sits on clutch (8-10 eggs). Any of the heavy or dual-purpose breeds are suitable for sitting and 2 years old are more reliable than pullets.

- ***The signs of broodiness are:***
- 1. Cessation of the egg lying.
- 2. Paleness or shrinking of comb and wattles.
- 3. Disinclination to leave the nest.
- 4. Characteristic clucking noise and ruffling of the feathers when disturbed.

Brooding

THE PRINCIPLE REQUIREMENTS FOR THE INCUBATING HEN ARE:

- 1. A plentiful supply of drinking water and food.
- 2. Protection against insect pests and vermin and
- 3.On hatching, protection of chick from predators.
- 4. The nest should be placed in some quite, cool corner at ground level.
- 5. Allow the hen to leave the nest only 10 minutes at morning and at evening. During that time offer her food, fresh water and sand bath.

B. Artificial incubators:

- **a. Choosing the right eggs.**
- 1. First, you should consider the chickens responsible for fertilizing and laying the eggs. The eggs should come from :
- A. healthy adult chickens that have a high fertility percentage, were not disturbed during mating season,
- B. were fed a good diet, and are not related to each other.

B. Artificial incubators:

- **a. Choosing the right eggs.**
- 2. Examine the eggs. They should also be :of a

A. regular shape and average size, not larger or smaller than a typical egg produced by your chickens.

B. Their shells should display no holes or cracks because such damage could encourage disease organisms to penetrate the egg.

C. you should not wash or wipe the egg, as you could remove the egg's natural protective coating that guards against such organisms.

B. Artificial incubators:

- **a. Choosing the right eggs.**
- 3. The eggs should also be cared for correctly before incubation. This will help ensure a productive hatch. To care for the eggs
  - Store eggs in a cool, humid area, ideally 55'F with 75% humidity.
  - Store eggs with the large end up.
  - Turn eggs daily, use an X and O marking system to keep track.
  - Eggs can be stored for up to seven days.
  - Allow cool eggs to warm up to the ambient temperature where the incubator is located before putting them into the incubator.

b. Selecting an incubator:

Chickens can either be hatched by using a setting hen or an incubator. There are two types of incubators: **forced-air and still-air**.

Incubator

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R-COM

Digital incubator

2.Maintaining the proper incubator conditions:

- Essentials must be taken during incubation.
- **1.Humidity:** The incubator must be situated in insulated room and with **55%** relative humidity.
- **2.Ventilation:** Sufficient oxygen in the circulation to supply the need of growing embryos by placing the incubator at the centre of the room.
- **3.Temperature** in small incubators about 60% R.H. is needed at 39 ^{o}C up to 18th day while in larger ones at 38 ^{o}C, 60% R.H. is required, but thereafter till chick is hatched up to 70% R.H. is needed.

Maintaining the proper incubator conditions:

4.Turning: Incubated eggs should be turned 5 times daily up to the 18th day and not less twice per day then transferred to hatchery, which important for:

- 1. prevention the adhering with chorioallantois.
- 2.keep centralization of the embryo.

Maintaining the proper incubator conditions:

- **5.Candling** by passing a beam of light through the eggs in dark place. The eggs candled at the 7th and 18th day to remove the infertile eggs and dead embryos eggs.
- A. Infertile eggs show up clear,
- B. Dead germs are seen as a dark spot of varying size.
- C. Alive embryos show a shadow from which blood vessels.
- D. Dead embryos do not make a uniform darkness but appear to have an air space at both ends of the egg.

Candling

D·M·P

LOWER HATCHABILITY MAY BE ATTRIBUTED TO:

- **1. Infertility:**
- a. Infertility attributable to cock as it too young or too old.
- b. Males of light breeds mate when only 6 months old, but full mating capacity is not reached till 8 months of age and decline after the age of 2 years.
- c. the maximum number of hens for each cock should be 10-12. Fertile eggs may be laid within 24 hr. after mating.
- d. The maximum number of fertile eggs is not laid until **7 days** after the cock has been put with the hens. When the cock is removed, fertile eggs continued up to **10 days** later, and there after fertile eggs reduced.

LOWER HATCHABILITY MAY BE ATTRIBUTED TO:

- **2. Deficiency of the diet:** Fertility severely influenced when cocks and hens for any length of time, fed a diet deficient in any of the essential constituent, also the chicks may be defective and die during incubation or soon after hatching.
- a. The vitamins are of special importance, Vit. A deficiency adversely affects the reproductive cells.
- b. A lack of Vitamin E results in the death of embryos about the 4th day of incubation
- c. Also deficiency of certain members of Vitamin B results in death or deformity of the embryos at different stages of development or poor change of survival for the hatched individuals.
- d. Deficiency or excess of certain minerals has almost equally bad result.

LOWER HATCHABILITY MAY BE ATTRIBUTED TO:

3. Damage of storage and faulty handling:

a. Eggs should remain fit for hatching up to 7 days when stored at or near the optimum temperature 12.8 °C.

b. If they kept higher or lower than 12.8 °C, the fertility and hatchability decreased.

c. During storage, eggs do not need turning periodically but they should be placed with the broad ends up. Much agitation during transit before incubation lowers hatchability but the effect is considerably reduced if the eggs allowed to stand at rest for at least 24 hr.

d. Eggs should remain at 32 ^{o}C for 3 days; 23.9 ^{o}C for 9 days; 15.6 ^{o}C for 25 days; 7 ^{o}C for 65 days and at 2.8 ^{o}C for 100 day Much agitation during transit before incubation lowers hatchability.

LOWER HATCHABILITY MAY BE ATTRIBUTED TO:

- **4.Physical defects of the eggs:** Obviously eggs of abnormal shape, weight (than the average of the flock) and those with soft shell or porous are not suitable for incubation, the facts that reduce hatchability. In the hot season the number of abnormal eggs increased.

- **5.Hereditary defects**:. There are at least about 20 lethal mutations have been recorded and probably much of the mal-positioning indirectly due to hereditary factors.

LOWER HATCHABILITY MAY BE ATTRIBUTED TO:

- **6. Diseases:** Several diseases are passed through the eggs but only a few affect either fertility or hatchability. The commonest disease is bacillary white diarrhea, which cause death of a certain number of infected chicks in the shell and great number not long far after hatching. Routinely detection and isolation of infected hens from the flock should be carried out periodically by rapid agglutination field test.

Hatching

Chicken Hatching Process

visuaiphotos.com

O.K

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Chick rearing:

- ***a. Natural brooding*:**
- the hen and chicks usually housed in a small group. The hen and chicks usually housed in a small group for 10 days after the chick is hatched. When the chicks are 5-6 weeks old the hen may be removed and subsequent artificial rearing of the chickens.

CHICK REARING:

- ***b. Artificial brooders:*** widely used now a day. The brooders may be either, permanent brooder house or portable one. The permanent brooder house contain sections, each holding 75-100 chicks or in separate small houses. The starting temperature should be 32.2 ^{o}C and gradually reduced each week until the end of 6-8 weeks.

- ***The chief points to notice are:***
- 1. Avoid over or under-heating as either can cause chills.
- 2. Avoid draughts.
- 3. Avoid over crowding (less than 100 ones)

Brooding house

CHICK SEXING:

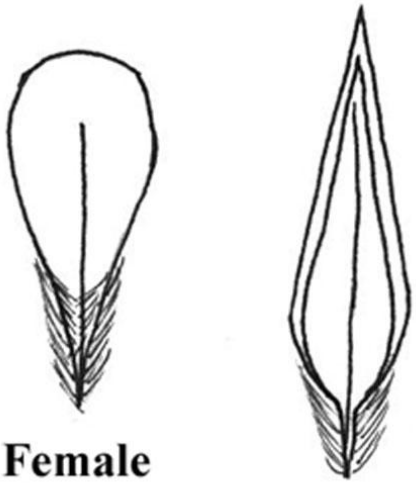
- This means determination the sex of day old chicks by:
- 1. Examination of the cloacae to differentiate between the male and female genital eminence.
- 2. In later stages from 2.5-3.5 month of age. The cockerel are more prominent than pullets More larger comb a wattles, earlier at about 9-12 weeks, cockerels are best indicated by the shape of the ends of the neck hackles and posterior saddle feathers, in which there are markedly narrow elongated and pointed, in the females they are blunt and rounded.
- 3. Self chick sexing: in some breeds at the day one old age the male has white feathers on his head not in the females.

Sexing

Male

Female

Waterfowl from dayold to four weeks

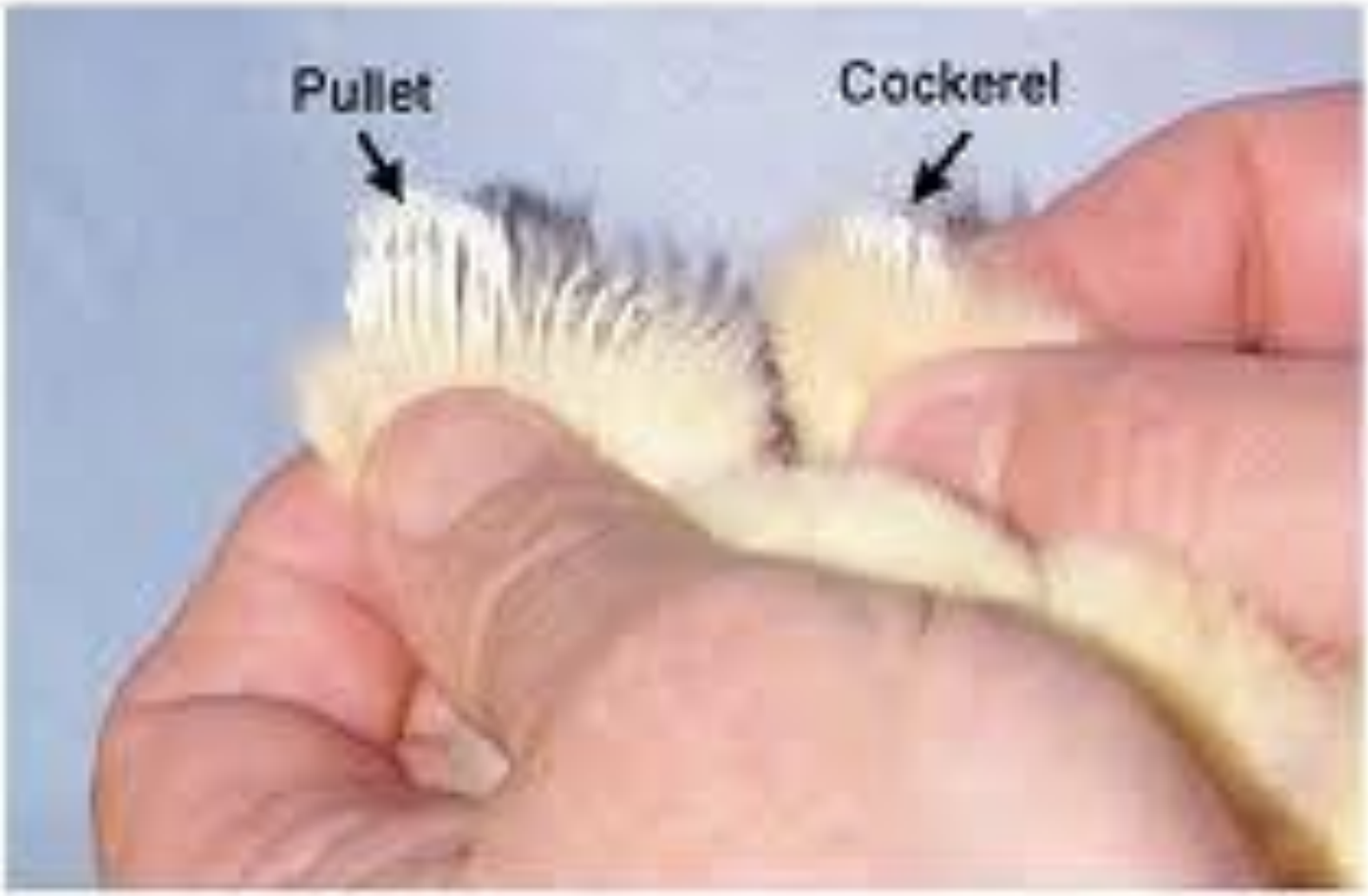


Female

Male

Poultry after eight weeks

Comparison of Pullet and Cockerel One Day Old Chicks



Pullet (?)
Cockerel (?)

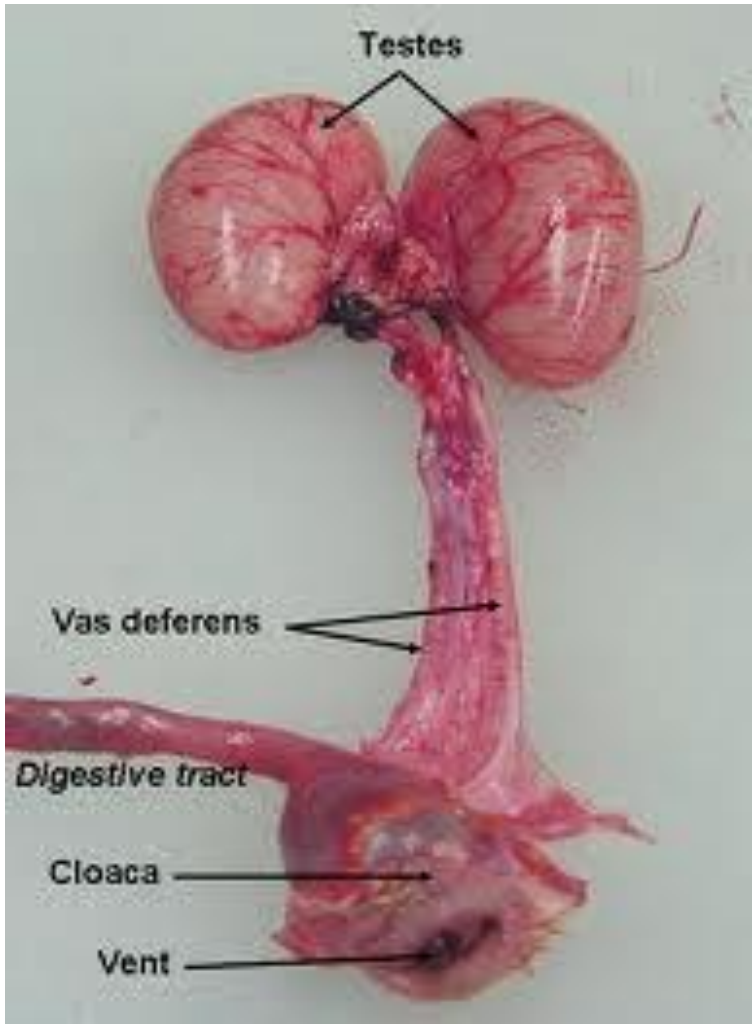
CAPONISATION:

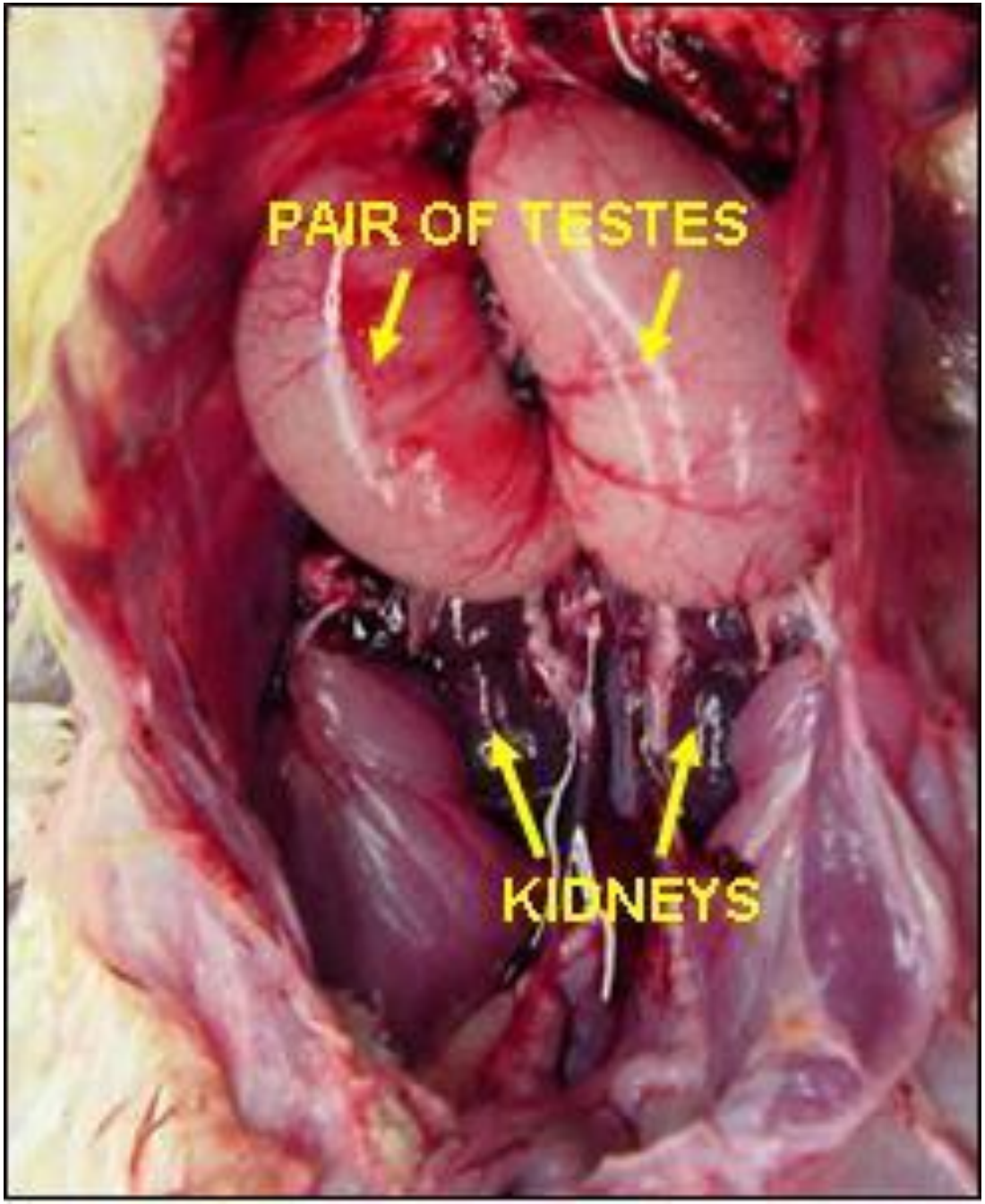
- Means the removal of the tests or male organs by:

1.Surgically

2.Chemically: by inserting a dried pellet of **Stelbesterol** in a skin pocket made in the back of the neck as near as the head as possible.

Caponization




PAIR OF TESTES
KIDNEYS

THANK YOU ...